RATING METHODOLOGY

12 August 2024

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Contacts

Dean Diaz +1.212.553.4332
Associate Managing Director
dean.diaz@moodys.com

CLIENT SERVICES

Americas	1-212-553-1653
Asia Pacific	852-3551-3077
Japan	81-3-5408-4100
EMEA	44-20-7772-5454

Rating Methodology

Environmental Services and Waste Management

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This methodology is no longer in effect. For more information on rating methodologies used by Moody's Investor Services, visit https://ratings.moodys.com/rating-methodologies.

This rating methodology replaces the *Environmental Services and Waste Management* methodology published on May 10, 2023. While this methodology reflects many of the same core principles as the 2023 methodology, we have made changes to the scorecard to provide greater consistency of approach across our corporate sector methodologies. Within the Leverage and Coverage factor, we have replaced the EBIT/Interest Expense sub-factor with EBITDA/Interest Expense and replaced the FFO/Debt sub-factor with RCF/Net Debt. We have changed some sub-factor scoring thresholds. We have also made limited editorial updates to some sections of the methodology.

Scope

This methodology applies to private-sector companies globally that are primarily engaged in providing environmental services, including industrial cleanup, nuclear decommissioning and disposal, freshwater and wastewater processing, and post-storm restoration. The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenues, earnings and cash flows.

This methodology also applies to companies globally that are primarily engaged in providing solid waste management services, including collection, transfer, disposal, and recycling services for residential, commercial, industrial and municipal customers. Companies rated using this methodology may also perform services in ancillary areas, such as hazardous waste disposal. The environmental services sub-sector includes all waste management other than solid waste management.

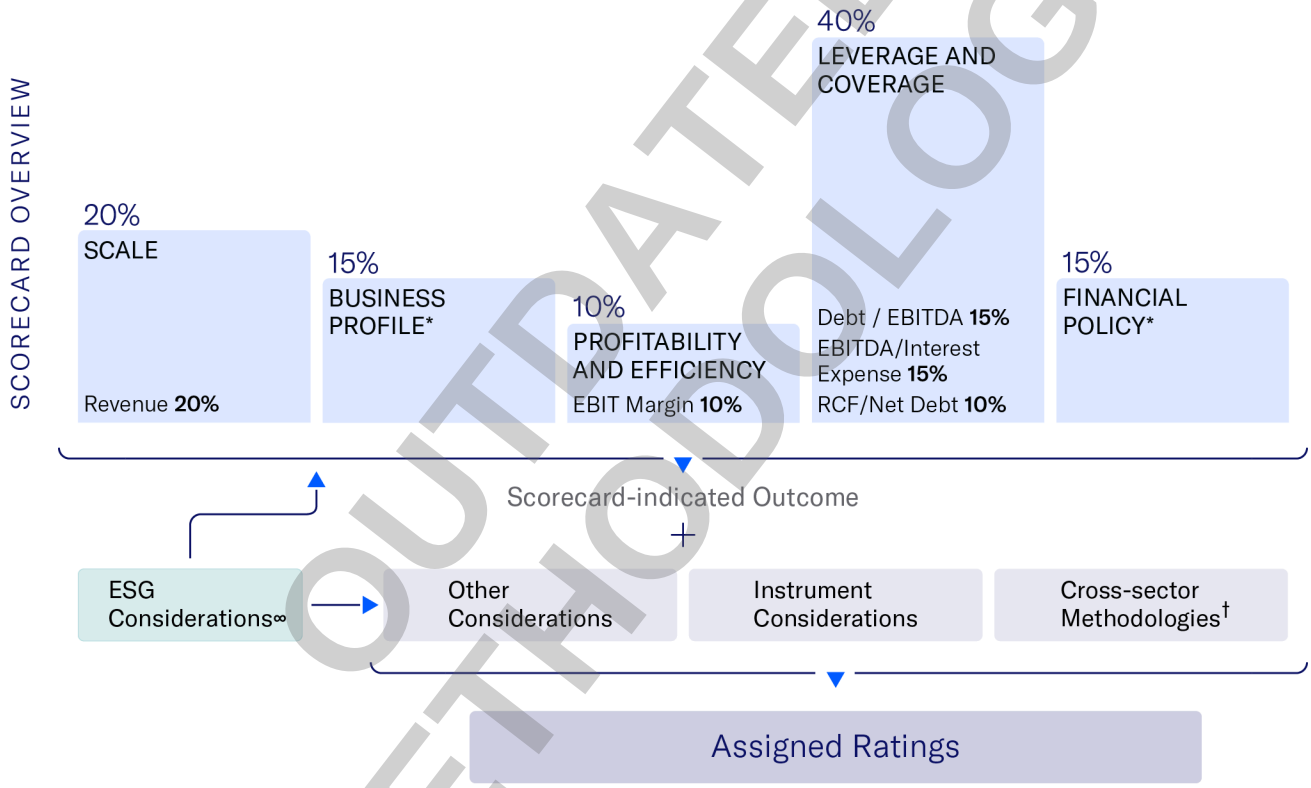
Rating approach

In this rating methodology, we explain our general approach to assessing credit risk of issuers in the environmental services and waste management industry globally, including the qualitative and quantitative factors that are likely to affect rating outcomes in this sector. We seek to incorporate all material credit considerations in ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

The following schematic illustrates our general framework for the analysis of environmental services and waste management companies, which includes the use of a scorecard. The scorecard-indicated outcome is not expected to match the actual rating for each company. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1

Illustration of the environmental services and waste management methodology framework



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* This factor has no sub-factors.

∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.

† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

Source: Moody's Ratings

Environmental services and waste management scorecard

For general information about how we use the scorecard and for a discussion of scorecard mechanics, please see the "Using the scorecard to arrive at a scorecard-indicated outcome" section. The scorecard does not include or address every factor that a rating committee may consider in assigning ratings in this sector. Please see the "Other considerations" and "Limitations" sections.

Exhibit 2

Environmental services and waste management scorecard

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Scale (20%)									
Revenue (USD Billion)[1]	20%	≥ $40	$20 - $40	$10 - $20	$4 - $10	$1.5 - $4	$0.5 - $1.5	$0.25 - $0.5	< $0.25
Factor: Business Profile (15%)									
Business Profile	15%	Environmental Services: Expected to have a very strong and sustainable business position across multiple major regions or countries and a broad range of environmental businesses. Extremely high barriers to entry.	Environmental Services: Expected to have a strong, sustainable business position across multiple major regions or countries and a range of environmental businesses. High barriers to entry.	Environmental Services: Expected to have a strong business position across multiple major regions or countries and a number of environmental businesses. Meaningful barriers to entry.	Environmental Services: Expected to have a solid business position across multiple large regions or countries and several environmental businesses. Moderate barriers to entry.	Environmental Services: Multi-region player with a competitive business position in several environmental businesses; revenues are somewhat concentrated. Some barriers to entry.	Environmental Services: Regional player with a competitive business position in at least two environmental businesses. Revenues are relatively concentrated. Low barriers to entry.	Environmental Services: Regional or local player with a somewhat competitive business position in one or two environmental businesses. Revenues are highly concentrated. No barriers to entry.	Environmental Services: Local player with no meaningful competitive advantage and a marginally profitable or loss-making business position in a narrow aspect of the environmental business.
		Solid Waste Management: Expected to have top market positions across multiple major countries or regions and services that include all elements of the waste stream. Has a sustainably high level of waste internalization.	Solid Waste Management: Expected to have leading market positions across multiple major countries or regions and services that include all elements of the waste stream. Has a sustainably high level of waste internalization.	Solid Waste Management: Expected to have leading market positions across several major countries or regions and services across a wide range of the waste stream. Some degree of revenue concentration may exist. Has a high level of waste internalization.	Solid Waste Management: Expected to have a sustainable market position in at least one major country or region and services across several elements of the waste stream. Single market revenue concentration may exist. Has a meaningful level of waste internalization.	Solid Waste Management: Multi-regional player offering services across several elements of the waste stream. Has a strong competitive position in its key markets. Revenues are relatively concentrated. Has a moderate level of waste internalization	Solid Waste Management: Regional player with services across at least two elements of the waste stream and a sustainable competitive position in several of its markets. Revenues are relatively concentrated. Has a modest level of waste internalization.	Solid Waste Management: Regional or local market player with services in one or two elements of the waste stream and a tenuous competitive position in its markets. Has a modest level of waste internalization.	Solid Waste Management: Local market player with a deteriorating competitive position. Service offerings limited to waste collection or hauling and no landfill.

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	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Profitability and Efficiency (10%)									
EBIT Margin[2]	10%	≥ 25%	20% - 25%	15% - 20%	10% - 15%	7% - 10%	4% - 7%	1% - 4%	< 1%
Factor: Leverage and Coverage (40%)									
Debt / EBITDA[3]	15%	≤ 0.5x	0.5x - 1x	1x - 2x	2x - 3x	3x - 4x	4x - 6x	6x - 8x	> 8x
EBITDA / Interest Expense[4]	15%	≥ 30x	20x - 30x	10x - 20x	7x - 10x	4x - 7x	2x - 4x	1x - 2x	< 1x
RCF / Net Debt[5]	10%	≥ 50%	35% - 50%	25% - 35%	15% - 25%	7% - 15%	0% - 7%	(5)% - 0%	< (5)%
Factor: Financial Policy (15%)									
Financial Policy	15%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to very strong credit profile over the long term.	Expected to have very conservative financial policies; (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions, or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions, or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

[1] For the linear scoring scale, the Aaa endpoint value is $60 billion. A value of $60 billion or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
[2] For the linear scoring scale, the Aaa endpoint value is 40%. A value of 40% or better equates to a numeric score of 0.5. The Ca endpoint value is 0%. A value of 0% or worse equates to a numeric score of 20.5.
[3] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The Ca endpoint value is 12x. A value of 12x or worse equates to a numeric score of 20.5, as does a negative Debt/EBITDA value.
[4] For the linear scoring scale, the Aaa endpoint value is 40x. A value of 40x or better equates to a numeric score of 0.5. The Ca endpoint value is 0x. A value of 0x or worse equates to a numeric score of 20.5
[5] For the linear scoring scale, when net debt is positive, the Aaa endpoint value is 75%. A value of 75% or better equates to a numeric score of 0.5. The Ca endpoint value is (10)%. A value of (10)% or worse equates to a numeric score of 20.5. When net debt is negative and RCF is positive, the numeric score is 0.5. When net debt is negative and RCF is negative or zero, the numeric score is 16.5.
Source: Moody's Ratings

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Sector overview

Environmental services companies provide services for various industries but they have a common need to optimize sizable asset bases, to build and utilize customer relationships, and to secure contracts that provide more predictable workflow. In most cases, intermediate to long-term revenue growth opportunities are underpinned by growing environmental awareness. However, short-term demand is subject to significant variability, particularly in markets that are fast-growing or highly sensitive to the overall pace of economic activity.

Solid waste management companies use similar business models and strategies for increasing returns on invested capital. Route density and waste internalization — where collected waste is disposed in company-owned landfills — provide efficiencies that usually support robust margins. Low product substitution and the fairly steady demand nature of waste creation dampen revenue volatility. Differences in geographic scope, aggregate size, and concentration in one or more of collection, transfer, disposal, recycling or other service are the primary operating differentiators among solid waste management companies.

Discussion of the scorecard factors

In this section, we explain our general approach for scoring each scorecard factor or sub-factor, and we describe why they are meaningful as credit indicators.

Factor: Scale (20% weight)

Why it matters

Scale is an important indicator of the overall depth of an environmental services or waste management company's business and its success in attracting a variety of customers, as well as its resilience to shocks, such as sudden shifts in demand or rapid cost increases.

Scale also can be an indicator of greater resilience to changes in product demand, geographic diversity, cost absorption, R&D capabilities and bargaining strength with customers and suppliers.

How we assess it for the scorecard

Scoring for this factor is based on one sub-factor: Revenue.

Revenue

Scale is measured (or estimated in the case of forward-looking expectations) using total reported revenue in billions of US dollars.

Factor: Business Profile (15% weight)

Why it matters

The business profile of an environmental services or waste management company is important because it greatly influences its ability to generate sustainable earnings and operating cash flows. Entry into this industry often requires significant investment, such as costs associated with developing landfill sites or building response capabilities for environmental cleanup projects. Large enterprises with more geographically diverse operations are better positioned to fund these investments and to achieve an adequate return on the investment. Moreover, companies with a more robust business profile, characterized by strong presence across multiple markets or geographies, are better able to weather the business cycle or other operating risks affecting the sector.

How we assess it for the scorecard

In assessing this factor, we differentiate between the environmental services sub-sector and the solid waste management sub-sector in recognition of the differences in their respective business models and keys to success. For clarity, the environmental services sub-sector includes all waste management other than solid waste management.

For environmental services companies, competition revolves around performance, price, substitution, and ability to perform specialized environmental services. Barriers to entry can be high in cases of unique assets.

For solid waste management companies, a broad range of waste disposal service elements across the waste stream and high waste internalization are key levers for sustaining market share and pricing power. Geographic scope, aggregate size and concentration in one or more of collection, transfer, disposal, recycling or other operations are the primary operating differentiators among solid waste management companies.

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Factor: Profitability and Efficiency (10% weight)

Why it matters

Profits matter because they are needed to generate sustainable cash flow and maintain a competitive position.

Although revenue volatility for solid waste management companies is typically lower than GDP volatility, financial flexibility is important because companies need adequate funds to be able to adapt to changes in market conditions.

This factor comprises one sub-factor.

EBIT Margin

The ratio of earnings before interest and taxes to revenue (EBIT Margin) is an indicator of a company's operating success and the effectiveness of management. Service offerings, company-specific operations and market-wide conditions can affect operating efficiency and profitability.

How we assess it for the scorecard

Scoring for this factor is based on one sub-factor: EBIT Margin.[1]

EBIT Margin

The numerator is EBIT, and the denominator is revenue.

Factor: Leverage and Coverage (40% weight)

Why it matters

Leverage and coverage measures provide important indications of how much financial risk an environmental services or waste management company is willing to undertake. These metrics are also important indicators of a company's financial flexibility, its ability to invest in growth opportunities and service debt.

This factor comprises three sub-factors:

Debt / EBITDA

The ratio of total debt to earnings before interest, taxes, depreciation and amortization (Debt/EBITDA) is an indicator of debt serviceability and financial leverage. The ratio is commonly used in this sector as a proxy for comparative financial strength.

EBITDA / Interest Expense

The ratio of earnings before interest, taxes, depreciation and amortization to interest expense (EBITDA/Interest Expense) is an indicator of a company's ability to meet its interest obligations.

RCF / Net Debt

The ratio of retained cash flow to net debt (RCF/Net Debt) is an indicator of a company's cash generation (before working capital movements and capital expenditures, and after dividend payments) relative to its net debt.

How we assess it for the scorecard

Scoring for this factor is based on three sub-factors: Debt/EBITDA; EBITDA/Interest Expense; and RCF/Net Debt.[2]

Debt / EBITDA

The numerator is total debt, and the denominator is EBITDA.

EBITDA / Interest Expense

The numerator is EBITDA, and the denominator is interest expense.

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RCF / Net Debt

This numerator is retained cash flow, and the denominator is net debt.

Factor: Financial Policy (15% weight)

Why it matters

Financial policy encompasses management and board tolerance for financial risk and commitment to a strong credit profile. It is an important rating determinant, because it directly affects debt levels, credit quality, the future direction for the company and the risk of adverse changes in financing and capital structure.

Financial risk tolerance serves as a guidepost to investment and capital allocation. An expectation that management will be committed to sustaining an improved credit profile is often necessary to support an upgrade. For example, we may not upgrade the ratings of a company that has built flexibility within its rating category if we believe the company will use that flexibility to fund a strategic acquisition, cash distribution to shareholders, spinoff or other leveraging transaction. Conversely, a company's credit rating may be better able to withstand a moderate leveraging event if management places a high priority on returning credit metrics to pre-transaction levels and has consistently demonstrated the commitment to do so through prior actions. Liquidity management is an important aspect of overall risk management and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which is discussed in the "Other considerations" section.

How we assess it for the scorecard

We assess the issuer's desired capital structure or targeted credit profile, its history of prior actions, including its track record of risk and liquidity management, and its adherence to its commitments. Attention is paid to management's operating performance and use of cash flow through different phases of economic and industry cycles. Also of interest is the way in which management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges or regulatory pressures. Considerations include a company's public commitments in this area, its track record of adhering to commitments and our views on the ability of the company to achieve its targets.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential negative impact of M&A or other types of balance-sheet-transforming events. Management's appetite for M&A activity is assessed, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. Frequency and materiality of acquisitions and previous financing choices are evaluated. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider negative repercussions caused by shareholders' willingness to sell the company.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. A track record of favoring shareholder returns at the expense of debtholders is likely to be viewed negatively in scoring this factor.

Other considerations

Ratings may reflect consideration of additional factors that are not in the scorecard, usually because the factor's credit importance varies widely among the issuers in the sector or because the factor may be important only under certain circumstances or for a subset of issuers. Such factors can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends can also affect ratings.

The following are examples of additional considerations that may be reflected in our ratings and that may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and sub-factors in the scorecard.

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Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given issuer is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, please see our methodology that describes our general principles for assessing ESG risks.[3]

Liquidity

Liquidity is an important rating consideration for all environmental services and waste management companies, and extremely weak liquidity can heavily affect ratings in many cases. However, the relative strength or weakness of liquidity may not have a substantial impact in discriminating between two issuers with an otherwise similar credit profile. Liquidity can be particularly important where companies have large short-term demands on liquidity, for example in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely near-term liquidity requirements from the perspective of both sources and uses of cash. For more information on our approach, please see our liquidity cross-sector methodology.[4]

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies, and philosophies and in evaluating management performance relative to performance of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in an issuer's fundamental creditworthiness, which may cause actual ratings to be lower than the scorecard-indicated outcome. Event risks — which are varied and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized firm. Some other types of event risks include M&A, asset sales, spinoffs, shareholder distributions, litigation, pandemics, significant cybercrime events and geopolitical conflicts.

Parental support

Ownership can provide ratings lift for a particular company in the environmental services and waste management sector if it is owned by a highly rated owner(s) and is viewed to be of strategic importance to those owners. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic incentives to provide support to the issuer in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or has already done so in the past. Conversely, if the parent puts a high dividend burden on the issuer, which in turn reduces its flexibility, the ratings would reflect this risk.

Government-related issuers may receive ratings uplift due to expected government support. However, for certain issuers, government ownership can have a negative impact on the underlying Baseline Credit Assessment.[5] For example, price controls, onerous taxation and high distributions can have a negative effect on an issuer's underlying credit profile.

Using the scorecard to arrive at a scorecard-indicated outcome

1. Measurement or estimation of factors in the scorecard

In the "Discussion of the scorecard factors" section, we explain our analytical approach for scoring each scorecard factor or sub-factor, and we describe why they are meaningful as credit indicators. When a factor comprises sub-factors, we score at the sub-factor level. Some factors do not have sub-factors, in which case we score at the factor level.

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The information used in assessing the sub-factors is generally found in or calculated from information in the company's financial statements or regulatory filings, derived from other observations or estimated by Moody's analysts. We may also incorporate non-public information.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends of a company's performance as well as for peer comparisons. Financial metrics,[6] unless otherwise indicated, are typically calculated based on an annual or other 12-month period. However, the factors in the scorecard can be assessed using various time periods. For example, rating committees may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics incorporate our standard adjustments[7] to income statement, cash flow statement and balance sheet amounts for items such as underfunded pension obligations and operating leases. We may also make other analytical adjustments that are specific to a particular company.

2. Mapping scorecard factors to a numeric score

After estimating or calculating each factor or sub-factor, each outcome is mapped to a broad Moody's rating category (Aaa, Aa, A, Baa, Ba, B, Caa or Ca, also called alpha categories) and to a numeric score.

Qualitative factors are scored based on the description by broad rating category in the scorecard. The numeric value of each alpha score is based on the scale below.

Exhibit 3

Numeric equivalents for qualitative factor and sub-factor scores

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

Quantitative factors are scored on a linear continuum. For each metric, the scorecard shows the range by alpha category. We use the scale below and linear interpolation to convert the metric, based on its placement within the scorecard range, to a numeric score, which may be a fraction. As a purely theoretical example, if there were a ratio of revenue to interest for which the Baa range was 50x to 100x, then the numeric score for an issuer with revenue/interest of 99x, relatively strong within this range, would score closer to 7.5, and an issuer with revenue/interest of 51x, relatively weak within this range, would score closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the value of the metric that constitutes the lowest possible numeric score, and the value that constitutes the highest possible numeric score).

Exhibit 4

Ranges of numeric equivalents for quantitative factor and sub-factor scores

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

3. Determining the overall scorecard-indicated outcome

The numeric score for each sub-factor (or each factor, when the factor has no sub-factors) is multiplied by the weight for that sub-factor (or factor), with the results then summed to produce an aggregate numeric score. The aggregate numeric score is then mapped back to a scorecard-indicated outcome based on the ranges in the table below.

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Exhibit 5
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x < 1.5	Aaa
1.5 ≤ x < 2.5	Aa1
2.5 ≤ x < 3.5	Aa2
3.5 ≤ x < 4.5	Aa3
4.5 ≤ x < 5.5	A1
5.5 ≤ x < 6.5	A2
6.5 ≤ x < 7.5	A3
7.5 ≤ x < 8.5	Baa1
8.5 ≤ x < 9.5	Baa2
9.5 ≤ x < 10.5	Baa3
10.5 ≤ x < 11.5	Ba1
11.5 ≤ x < 12.5	Ba2
12.5 ≤ x < 13.5	Ba3
13.5 ≤ x < 14.5	B1
14.5 ≤ x < 15.5	B2
15.5 ≤ x < 16.5	B3
16.5 ≤ x < 17.5	Caa1
17.5 ≤ x < 18.5	Caa2
18.5 ≤ x < 19.5	Caa3
x ≥ 19.5	Ca

Source: Moody's Ratings

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For example, an issuer with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the corporate family rating (CFR) for speculative-grade issuers and to the senior unsecured rating for investment-grade issuers. For issuers that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or Baseline Credit Assessment for comparison to the scorecard-indicated outcome. For an explanation of the Baseline Credit Assessment, please refer to *Rating Symbols and Definitions* and to our cross-sector methodology for government-related issuers.[8]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a CFR to speculative-grade issuers or a senior unsecured rating for investment-grade issuers. For issuers that benefit from rating uplift from government ownership, we may assign a Baseline Credit Assessment.[9]

Individual debt instrument ratings may be notched up or down from the CFR or the senior unsecured rating to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. The documents that provide broad guidance for such notching decisions are the rating methodology on loss given default for speculative-grade nonfinancial companies, the methodology for notching corporate instrument ratings based on differences in security and priority of claim, and the methodology for assigning short-term ratings.[10]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, please see *Rating Symbols and Definitions*.[11]

Limitations

In the preceding sections, we have discussed the scorecard factors and many of the other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall rating methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to actual ratings.

The scorecard in this rating methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings to these companies. Credit loss and recovery considerations, which are typically more important as an issuer gets closer to default, may not be fully captured in the scorecard. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for issuers at the upper and lower ends of the rating scale.

The weights for each factor and sub-factor in the scorecard represent an approximation of their importance for rating decisions across the sector, but the actual importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors that are outside the scorecard, including those discussed above in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[12] Examples of such considerations include the following: how sovereign credit quality affects non-sovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking time periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology document does not include an exhaustive description of all factors that we may consider in assigning ratings in this sector. Companies in the sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for an issuer's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

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Moody's related publications

Credit ratings are primarily determined through the application of sector credit rating methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

For data summarizing the historical robustness and predictive power of credit ratings, please click here.

For further information, please refer to *Rating Symbols and Definitions*, which is available here.

Moody's Basic Definitions for Credit Statistics (User's Guide) can be found here.

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Endnotes

1 For definitions of our most common metrics, see *Moody's Basic Definitions for Credit Statistics (User's Guide)*. A link can be found in the "Moody's related publications" section.

2 For definitions of our most common metrics, see *Moody's Basic Definitions for Credit Statistics (User's Guide)*. A link can be found in the "Moody's related publications" section.

3 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

4 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

5 For an explanation of the Baseline Credit Assessment, please refer to *Rating Symbols and Definitions* and to our cross-sector methodology for government-related issuers. A link to a list of our sector and cross-sector methodologies and a link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

6 For definitions of our most common metrics, see *Moody's Basic Definitions for Credit Statistics (User's Guide)*. A link can be found in the "Moody's related publications" section.

7 For an explanation of our standard adjustments, see the cross-sector methodology that describes our financial statement adjustments in the analysis of nonfinancial corporations. A link can be found in the "Moody's related publications" section.

8 A link to a list of our sector and cross-sector methodologies and a link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

9 For an explanation of the Baseline Credit Assessment, please refer to *Rating Symbols and Definitions* and to our cross-sector methodology for government-related issuers. A link to a list of our sector and cross-sector methodologies and a link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

10 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

11 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

12 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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REPORT NUMBER 1392145

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Authors

Tomas O'Loughlin
Dean Diaz
Eric Greaser

CLIENT SERVICES

Americas	1-212-553-1653
Asia Pacific	852-3551-3077
Japan	81-3-5408-4100
EMEA	44-20-7772-5454

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